EWB COURSES

AI-Powered Cybersecurity

Assessment 01: Security Foundations

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Introduction

This assignment is about basic cybersecurity. It covers how to protect a student account, identify a suspicious email, understand a sign-in alert, and check whether an AI answer gives safe advice. The examples in this assignment are fictional and are used only for learning.

Task A – Secure a Student Account

A student named Mira uses the same short password for her email and learning portal. She sometimes approves unexpected sign-in requests, delays updates, keeps only one copy of her project, and allows a friend to use her laptop while she is signed in.

A1. Five risks and how to reduce them

Risk	Possible problem	Basic safety step
1. Reusing a short password	If someone finds the password, they may access both email and the learning portal.	Use a long, different password for every account. Turn on MFA if available.
2. Approving unknown sign-in requests	Another person may be trying to enter the account.	Deny unexpected requests. Open the official website yourself and report repeated prompts.
3. Delaying software updates	Old software may have security weaknesses that attackers can use.	Install phone, laptop, browser and app updates regularly.
4. Keeping only one project copy	A laptop problem, theft or ransomware could remove the only copy.	Keep another copy in trusted cloud storage or on a separate USB drive.
5. Sharing a laptop while signed in	A friend may read private information or change and submit work using Mira's account.	Lock the screen, sign out before sharing, and create a separate user account for guests.

A2. Which actions should Mira do first?

1. Change the reused password first because one password should not open more than one account.

2. Stop approving unknown MFA requests. She should deny them and inform the college if they continue.

3. Stop sharing the laptop while she is signed in. She should lock the screen or sign out before giving it to someone.

She should also install updates and make backup copies as soon as possible.

A3. Authentication and authorisation

Authentication means checking who a person is. For example, Mira enters her username, password and an MFA code to log in.

Authorisation means deciding what the logged-in person is allowed to do. Mira can view her own marks and submit her assignment, but she should not be able to change another student's marks.

In simple words, authentication checks the user, while authorisation checks what the user can do.

Task B – Analyse a Suspicious Message

The sample email says that a scholarship will expire soon. It asks the student to enter a college password and OTP, send an identity-card photo, and not contact the college office.

B1. Five warning signs

Warning sign	Why it is suspicious
The sender address is campus-awards.example.	It is different from the college domain campus.example.
The Reply-To address is help@personal-awards.example.	Replies would go to a different address.
The subject says “FINAL WARNING” and gives only 20 minutes.	It creates pressure and may stop the student from checking properly.
It asks for a password and OTP.	A scholarship office should not ask for these login secrets.
It says not to contact the college and asks for an ID-card photo.	It prevents independent checking and may collect personal information.

B2. How to check whether the message is real

1. Do not click the link or reply to the email.

2. Open the college website separately by typing its known address or using a trusted bookmark.

3. Contact the scholarship office using the phone number or email address shown on the official college website.

4. If the message is fake, report it to the college IT team and delete it.

HTTPS does not prove that a website is genuine. It only protects the connection to that website. Also, a sender name such as “Scholarship Desk” can be chosen by anyone, so the actual email address must be checked.

B3. What if the student already entered a password and OTP?

The student should open the official college portal using a trusted device and change the password immediately. They should sign out of other sessions if that option is available. If the same password is used for email, that password should also be changed. The student should deny any new sign-in requests and report the incident to the college IT team. They should share the time and details of the message, but never share the password or OTP.

B4. What cannot be confirmed from the email alone?

The email text alone cannot prove who owns the website or whether anyone has already used the link. The college IT team would need to check the full email details and relevant system records.

Task C – Triage a Sign-in Alert

The log is dated 18 October 2026. The usual classroom network is 192.0.2.10. Another sign-in attempt came from 198.51.100.23. The different IP address alone does not prove that an attacker was involved.

C1. Timeline of events

Time	What happened	Simple meaning
09:00–09:01	Password success and MFA approved from 192.0.2.10.	This looks like the student's normal classroom login.
10:14–10:15	Two password failures from 198.51.100.23.	Someone or something tried the password from another network.
10:16–10:18	Password success, MFA denied, then MFA approved from the same IP.	The correct password was entered and an MFA request was later approved.
10:20	A session was created from 198.51.100.23.	A successful session was created after the MFA approval.

C2. Possible explanation and another possibility

Likely explanation: Someone may have known the student's password and tried to sign in. The student said they approved a prompt because they wanted the repeated requests to stop. This may be an example of MFA fatigue.

Another possibility: The student may have been using another network, such as mobile data or a VPN. They may have entered the password incorrectly at first and then approved the request by mistake.

The IP address does not identify a person. The IT team can compare the device details and check how many MFA prompts were sent. They can also ask where the student was at that time.

C3. Three actions for the college IT team

1. Sign out the suspicious session and ask the student to reset the password and MFA settings through the official college process.

2. Save the sign-in logs and the student's statement for further checking. The logs should not be deleted.

3. Explain that unexpected MFA requests should be denied and reported. The IT team can also check whether the same IP was used for other accounts.

C4. Short incident summary

On 18 October 2026, student07 logged in normally at 09:00 from the usual classroom network. Later, there were two password failures and then a password success from another IP address. An MFA request was denied and a later request was approved at 10:18. A session was created at 10:20. The student said they approved the request to stop repeated notifications. This may be an unauthorised login, but the IP address does not prove who used it. The account should be secured and the logs should be saved.

Task D – Review an AI Security Answer

The sample AI answer says that the email is safe because it uses HTTPS, that the IP proves someone stole the password, that MFA prevents every attack, that suspicious logs should be deleted, and that the student's password should be sent to support.

D1. Five incorrect or unsafe statements

AI statement	Why it is wrong	Correct advice
The email is safe because it uses HTTPS.	HTTPS does not prove that the website belongs to the college.	Check the website address and contact the college through its official website.
The IP proves someone stole the password.	An IP address does not identify a person. The logs only show that the correct password was entered.	Treat it as suspicious and investigate other evidence.
MFA prevents every account attack.	MFA reduces risk, but a person can still approve a fake or unexpected request.	Use MFA carefully and deny requests that were not started by the user.
Delete the suspicious logs.	Deleting logs removes useful evidence.	Save the logs so the IT team can investigate.
Send the password to support.	Passwords should not be shared with anyone.	The IT team can reset the password without asking for the old one.

D2. A better prompt to give an AI tool

“Read only the fictional information I provide. Separate facts from guesses. Explain what is not known. Give safe steps that a student or college IT team can take. Do not ask for passwords, OTPs or identity documents. Do not say that an IP address identifies a person. Do not say that HTTPS proves a website is genuine. If there is not enough information, clearly say so.”

D3. Corrected security advice

The scholarship email should be treated as suspicious. Do not click its link, reply to it or send an identity-card photo. Check the scholarship only through the official college website or office. HTTPS does not prove that the sender is genuine. If a password and OTP were already entered, change the password using the official portal, sign out of other sessions and report the incident to college IT. In the sign-in log, a session was created from another IP after an MFA approval. This needs urgent checking, but the IP does not prove who was using it. Keep the logs and never send passwords to support.

D4. AI-use declaration

AI tools were used to help organise and prepare this assignment. The information was checked against the fictional assessment brief and the references below. No real passwords, OTPs, identity documents or live websites were used.

References

1. CISA – Secure Our World: https://www.cisa.gov/secure-our-world

2. CISA – Recognize and Report Phishing: https://www.cisa.gov/secure-our-world/recognize-and-report-phishing

3. CISA – Use Strong Passwords: https://www.cisa.gov/secure-our-world/use-strong-passwords

4. CISA – Turn on MFA: https://www.cisa.gov/secure-our-world/turn-mfa

5. CISA – Update Software: https://www.cisa.gov/secure-our-world/update-software

6. NIST (2024). Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile (NIST AI 600-1). https://doi.org/10.6028/NIST.AI.600-1

7. EWB Courses (18 October 2026). Assessment 01: Security Foundations – assignment brief.

End of Assignment